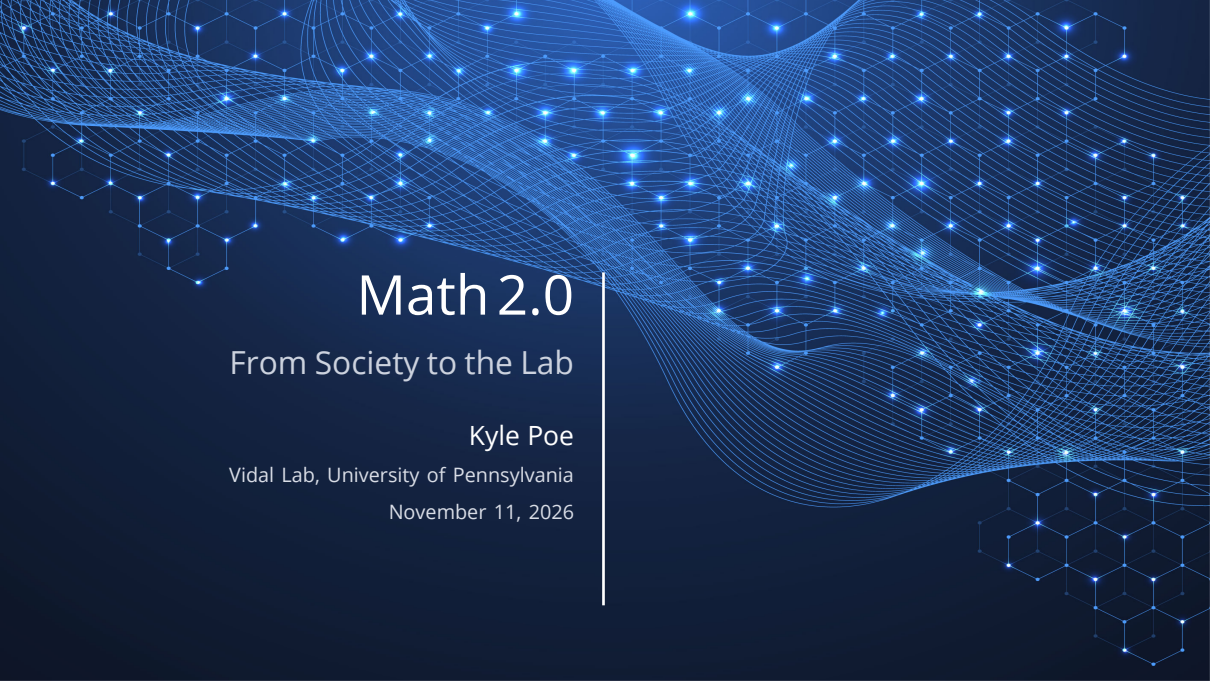


# Math 2.0

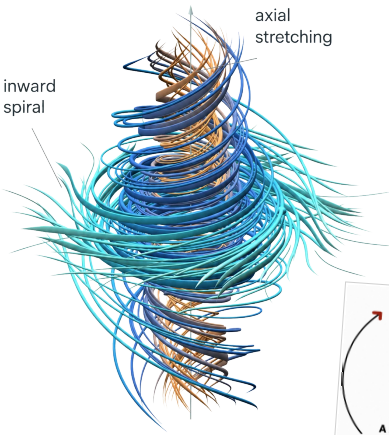
## From Society to the Lab

Kyle Poe

Vidal Lab, University of Pennsylvania

November 11, 2026

# September 8, 2026: Math changes forever




**OpenAI** @OpenAI

We're sharing a solution to the Navier-Stokes Millennium Prize Problem, one of the deepest problems at the frontier of mathematics.

nature

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nature › news › article

NEWS | 08 September 2026

**OpenAI claims huge maths**
**gh on a famed**
**n Problem'**

cked the physics of fluids
in mathematics.

**OpenAI Claims to Have Solved a $1M Math Problem: A Credit Dispute Has Already Begun**

Ten thousand AI agents resolved one of math's oldest open problems in 88 hours. A credit dispute over authorship erupted almost immediately.

BY TECHNOLOGY DESK · SEPTEMBER 9, 2026 · 3 MINS READ

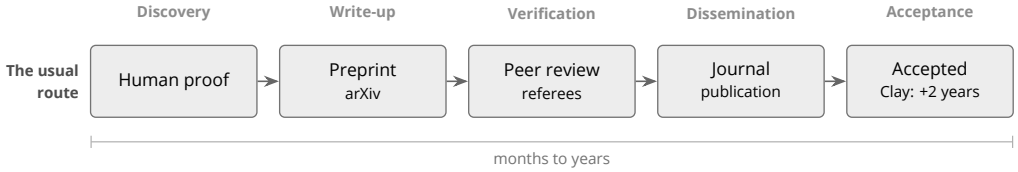



**OpenAI Cracks 200-Year-Old 'Millennium' Math Problem—But Credit Dispute Looms**

An NYU math professor claims the AI behemoth rushed to solve the problem after discovering his progress.

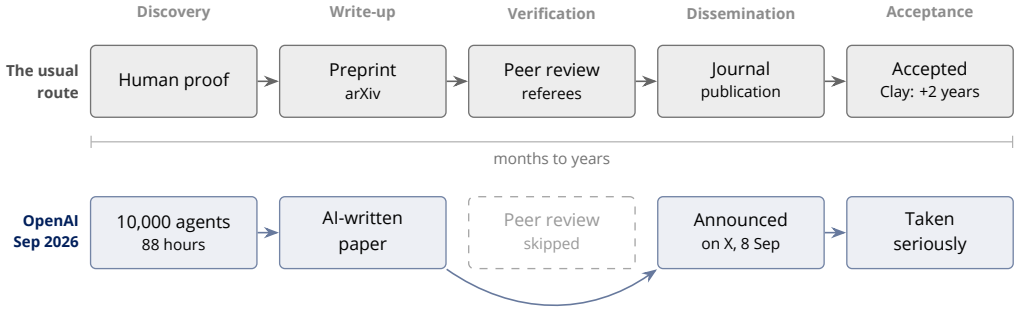
Sources: OpenAI on X, 8 Sep 2026 • Nature news, 8 Sep 2026 • Technology Desk, 9 Sep 2026 • Forbes • Visualization: OpenAI • Meme: @MozeTech on X

# Why believe it?



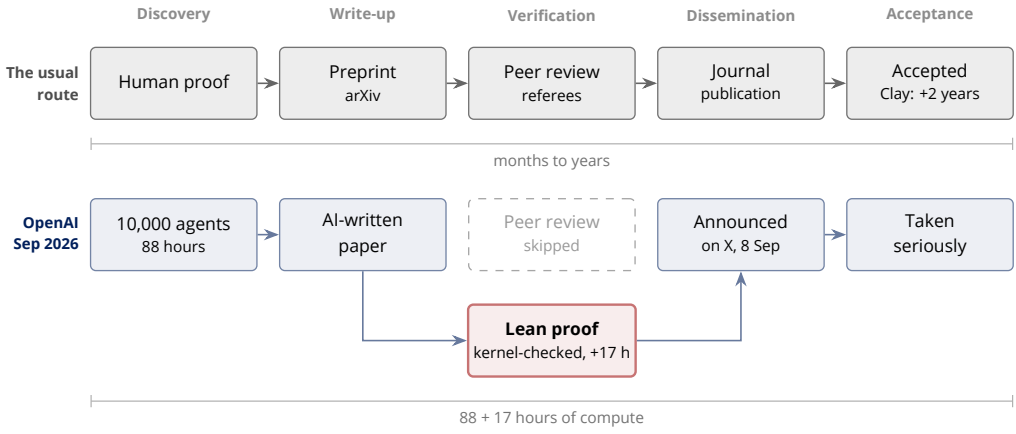

Sources: OpenAI, 8 Sep 2026 • Reisinger et al., *The Mathematical Sciences in an AI-enabled World*, AcadMathSci, Oct 2026 • Clay Mathematics Institute, Millennium Prize rules

# Why believe it?




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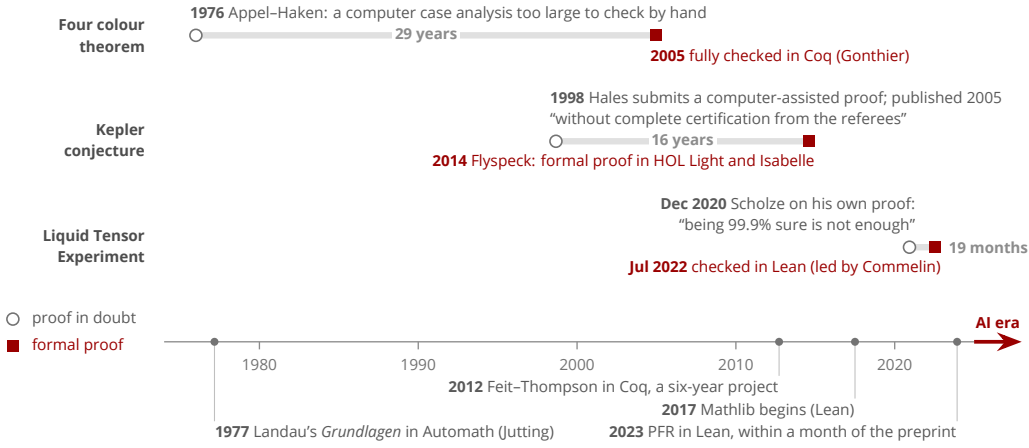
# Why believe it?




Lean verification, not merely OpenAI's reputation, made the community take this seriously.

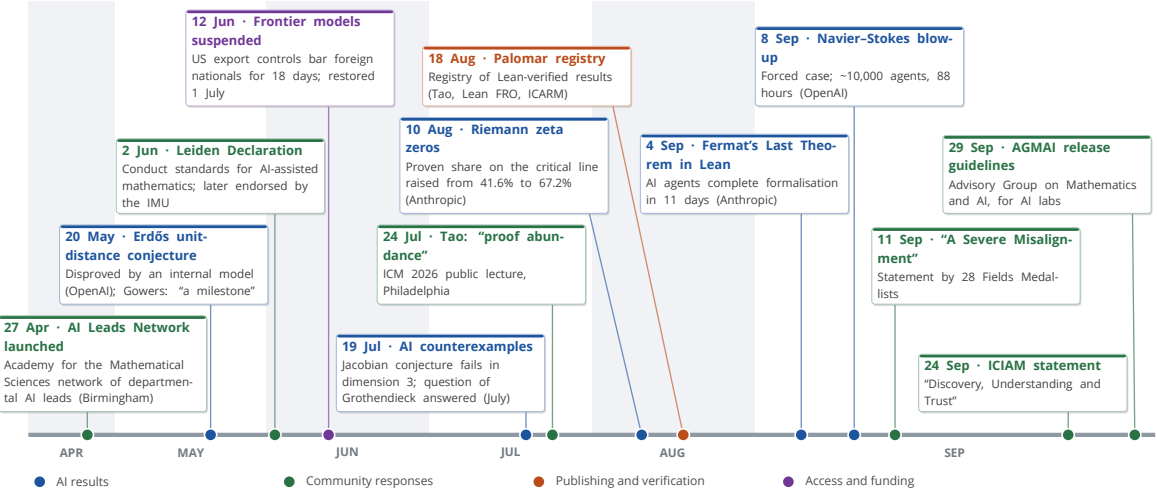
Sources: OpenAI, 8 Sep 2026 • Reisinger et al., *The Mathematical Sciences in an AI-enabled World*, AcadMathSci, Oct 2026 • Clay Mathematics Institute, Millennium Prize rules

# Timeline: Formalization in the Pre-AI Era



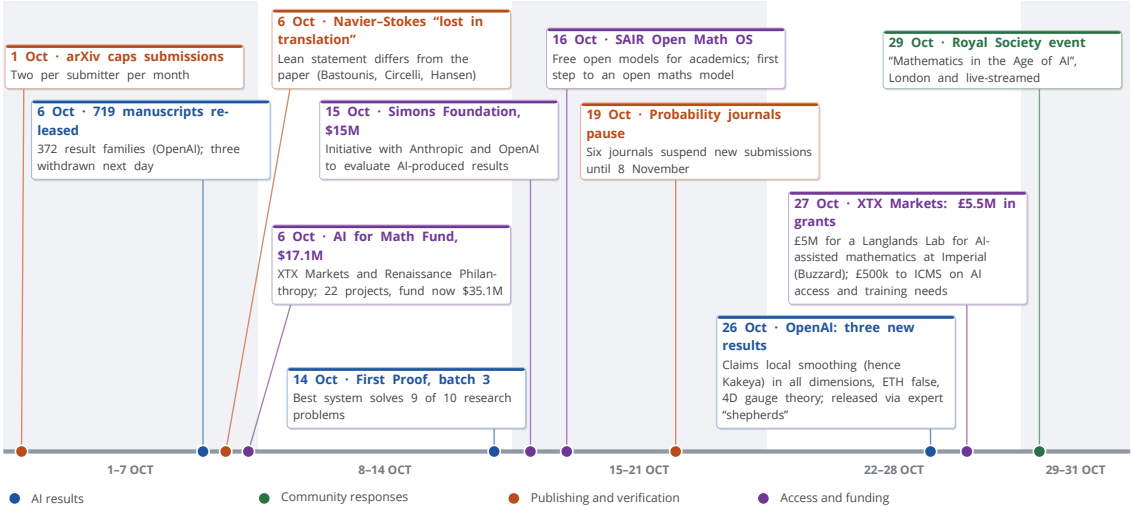

Sources: Jutting, thesis (1977) • Gonthier, MSR report (2005) • Hales, Annals 162 (2005); Hales et al., arXiv:1501.02155; Flyspeck repo • Gonthier et al., ITP 2013 • Scholze, Xena blog (Dec 2020); Lean blog (Jul 2022) • mathlib Community, CPP 2020 • Gowers et al., arXiv:2311.05762; Tao, blog (Dec 2023)

# Timeline: AI Transforms the Mathematical Sciences




Adapted from Reisinger et al., *The Mathematical Sciences in an AI-enabled World*, AcadMathSci, pre-publication draft, Oct 2026, Fig. 1 (April–September 2026)

# Timeline: AI Transforms the Mathematical Sciences




Adapted from Reisinger et al., *The Mathematical Sciences in an AI-enabled World*, AcadMathSci, pre-publication draft, Oct 2026, Fig. 1 (October 2026)

# Math 1.0 and Math 2.0

| | Light AI use | Heavy AI use |
|---|---|---|
| Weak human understanding | **Pre-math** | **Misaligned math** |
| Strong human understanding | **Math 1.0** | **Math 2.0** |



> *[Rather than blindly using AI to generate misaligned mathematics,] the more sustainable* "Math 2.0" *use of AI would be to strengthen all the ways in which human understanding interacts with mathematics.*
>
> — Terence Tao, *Machine Assisted Proof*, AIM Public Lecture, Caltech, 9 Oct 2026

*For mathematicians, theorists, and theoretically-inclined experimentalists alike, gracefully and ethically navigating the transition to Math 2.0 will require reckoning with societal-level issues as well as developing individual familiarity with the emergent paradigm of AI-assisted formalization.*

Quote and diagram: T. Tao, *Machine Assisted Proof*, AIM Public Lecture Series, Caltech, 9 Oct 2026 (caltech.edu/campus-life-events/calendar/aim-public-lecture-series-terence-tao-1)

# The rest of this talk

PART I
## Society

- Rampant AI use
- Community response
- What formalization fixes

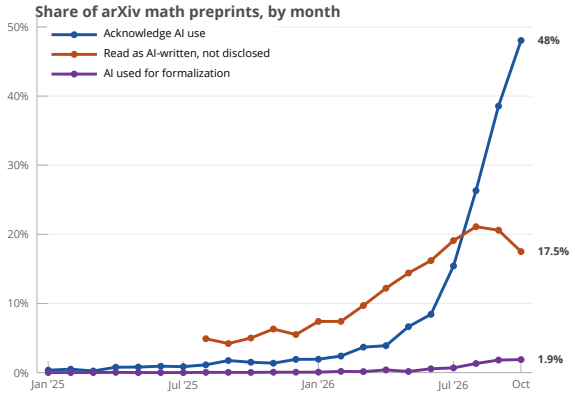
PART II
## The Lab

- In our field
- How it is being done
- What we can do

Part I

# Society

# The Poison Fountain




1. **Use is exploding.** Disclosed AI use went from 0.4% to 48% of math preprints in 21 months, doubling about every three months. Math submissions are up 75% on a year earlier (Sep 2026).

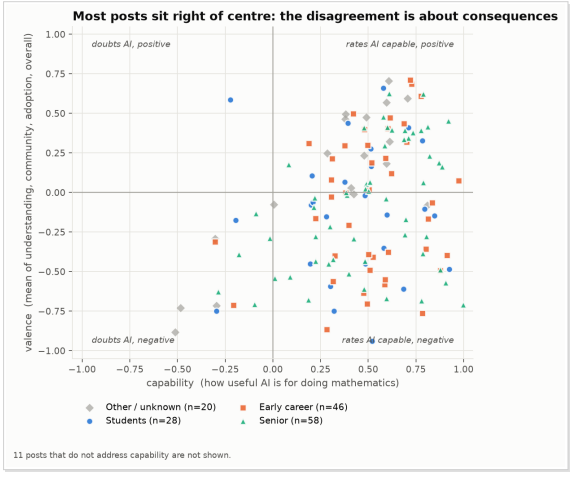
2. **Much of it is undisclosed.** About 1 in 6 math preprints read as AI-written without saying so. At ICLR 2026, 9% of submissions were mostly AI-written.

3. **Few certify.** OpenAI shipped Lean proofs with 42% of its 719 manuscripts. Across arXiv math, 1.9% acknowledge AI for formalization; in stat.ML, 0.4% even mention Lean.

4. **Review is overwhelmed.** 21% of ICLR 2026 reviews were fully AI-generated. arXiv capped submissions (1 Oct); six probability journals paused (19 Oct – 8 Nov).

Epoch AI (CC BY 4.0, 9 Nov 2026) • K. Martin (3 Nov 2026) • our arXiv count • Pangram, ICLR 2026 (self-reported) • AcadMathSci, Oct 2026

# Proofs and Prompts: A Sentiment Analysis




152 essays on *proofsandprompts.com*, each scored by an LLM

**89%**

rate AI as capable of doing mathematics

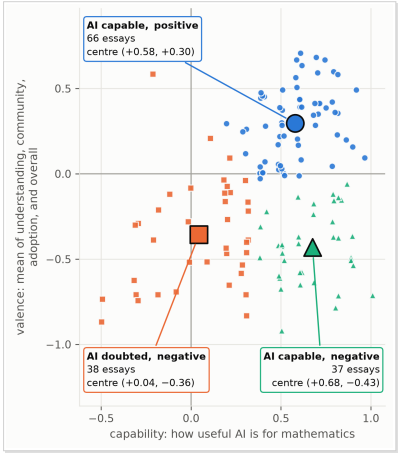
**49%**

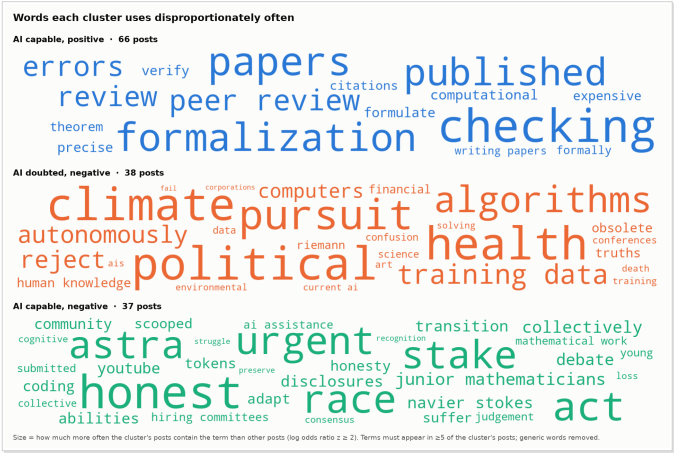
view AI as a negative force overall

**The field agrees that it works.**
**It disagrees about the consequences.**

Proofs & Prompts corpus (proofsandprompts.com), scored with Claude; analysis set of 152 posts, 141 addressing capability. Self-selected essays: an illustration, not a survey

# Proofs and Prompts: What Each Cluster Writes About




Proofs & Prompts corpus; scores from Claude (rubric.py); clustering and word statistics from cluster.py. Self-selected essays: an illustration, not a survey

# Credit, disclosure and confidentiality

| Problem | Case | Response | Status |
|---|---|---|---|
| **Credit** | AI reuses known ideas without attribution: 8 of Aletheia's 13 meaningful Erdős results were already in the literature. Kolda: "Human–AI scrapple" | Leiden Declaration (Jun 2026, endorsed by the IMU): human responsibility, attribution | • partial |
| **Priority** | Navier–Stokes priority dispute (Córdoba & Martínez-Zoroa; Buckmaster & Alpöge) | AGMAI release guidelines (29 Sep): review before announcement, release prompts. Labs withheld prompts | • open |
| **Disclosure** | 1 in 6 math preprints read as AI-written without saying so. At ICML 2026, 22.5% of reviewers used an LLM under a ban | Disclosure statements (arXiv, Springer, Leiden); SIAM AI policy v2.0 | • partial |
| **Confidentiality** | Referees uploading manuscripts to AI tools | LMS: misconduct. SIAM v2.0: "non-delegation of scholarly judgement" | • partial |

The rules already bind individuals; none of them can be checked from the paper alone.

Reisinger et al., AcadMathSci, Oct 2026 (A.3, C.1, C.4, C.5) • Kolda, Silicon Reckoner, Feb 2026 • Martin, Nov 2026 • Kim et al., arXiv 2609.19420

# Review capacity

| Problem | Case | Response | Status |
|---|---|---|---|
| **Volume** | arXiv math: 7,428 submissions in Sep 2026, against 4,252 a year earlier (+75%). A few prolific authors drive much of it | arXiv: two new submissions per submitter per month (1 Oct). STOC 2027: at most five per author | • in place |
| **Referees** | Six probability journals paused new submissions (19 Oct – 8 Nov). Editors report they cannot find a referee | Pauses; proposals for paid, AI-assisted refereeing (Maltsev) | • open |
| **AI reviewing** | ICLR 2026: 21% of reviews fully AI-generated | ICLR plans automated screening of reviews and submissions | • partial |
| **Bypassed review** | Results announced on X and GitHub before any review (Navier–Stokes; 719 manuscripts in a day) | Leiden and AGMAI: announce through peer review. Palomar registry of Lean-verified results (18 Aug); a proposed journal requiring Lean proofs (Guo, 17 Oct) | • partial |

Restricting and disclosing are in place; verifying is the structural response, and it is still a prototype.

AcadMathSci, Oct 2026 (§1.2) • Nee, Zhong & Ni, arXiv 2609.04872 • Fortnow, Sep 2026 • Pangram, ICLR 2026 (self-reported) • Tao, blog, 18 Aug and 17 Oct 2026

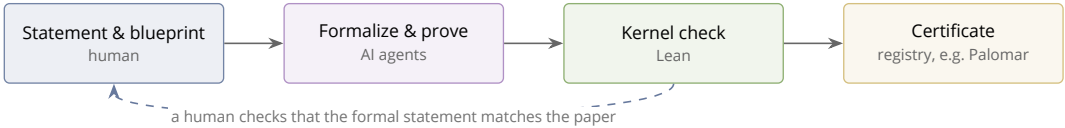
# Training, access and narrowing

| Problem | Case | Response | Status |
|---|---|---|---|
| **Training** | Students given unrestricted GPT-4 scored 17% lower once access was removed; a tutor with guardrails largely avoided the harm (Bastani et al.) | Vivas and in-person exams; AI-permitted work graded on formulation and validation | • partial |
| **Access** | Only 11% of researchers have institutional funding for their main AI tool (Springer Nature, 10,480 researchers). Navier–Stokes cost an estimated £4.9M | Token costs eligible on grants (proposed); SAIR Open Math OS: free open models for academics | • partial |
| **Narrowing** | AI adoption shrinks topic range by 4.6% and engagement by 22% (Hao et al., 41M papers). Agendas drift toward “problems easily amenable to current AI” | Reward understanding and long-term contribution over volume (AcadMathSci) | • open |

Reisinger et al., AcadMathSci, Oct 2026 (§1.6, §3.2, §3.3, §4.3, B.1) • Bastani et al., PNAS • Hao et al., arXiv 2412.07727

# AI-assisted formalization

AI translates mathematics into a proof assistant such as Lean and fills in the proofs. A small trusted kernel checks every step. Humans write, or review, what is being proved.




## What it addresses

- **Review capacity.** Referees review a page of statements, not 60 pages of proof.
- **Undisclosed AI.** The check does not depend on who, or what, wrote the proof.
- **Bypassed review.** Verify and register a result before announcing it.
- **Credit.** Registries record what produced each result.

## What stays open

- **Statement fidelity.** Compiling is not saying the same thing (Navier–Stokes).
- **Understanding.** Millions of machine-written lines explain nothing.
- **Significance and ideas.** Neither is a formal property.
- **Access and coverage.** Frontier compute; thin libraries in PDE and applied probability.

Tao, *Palomar*, blog, 18 Aug 2026 • Bastounis, Circelli & Hansen, arXiv 2610.08144 • research notes 01–03

# The paper and the code disagree

**Navier–Stokes** (Bastounis, Circelli & Hansen, Oct 2026)

- The paper's estimate (8.19) needs $m+4$ derivatives; the matching Lean theorem assumes $m+5$, and the pattern repeats.
- The paper's pressure bound depends on one quantity; the Lean bound depends on two, and is proved by a different route.
- They do not claim the paper is wrong.

**OpenAI's 719 manuscripts** (community audit of 245 formalized families)

- About 210 consistent at the level of statements; no theorem shown false.
- 21 headline results marked "(Lean)" were not formalized; 5 formalized only a supporting lemma.

> *Lean acceptance (compilation) of translation does not imply semantic preservation.*
>
> — Bastounis, Circelli & Hansen, arXiv 2610.08144

A certificate proves *a* theorem. Whether it is *the* theorem is a human question.

Bastounis, Circelli & Hansen, arXiv 2610.08144 • borism/openai-math-slop, issue #2 • research note 01, §5

# Three layers of checking

**1. Kernel**

**Does Lean accept the proof?** Cheap and mechanical. Watch for `sorry`, extra axioms and kernel bugs. • in place

**2. Statement**

**Does the formal statement say what the paper claims?** Scarce: "will never be solved purely in software" (Kontorovich). Navier–Stokes drifted here. • open

**3. Understanding**

**Does anyone understand the proof, and does it match the written one?** Scarce. Machine proofs run to millions of lines. • open

Layer 1 is solved. Layers 2 and 3 are the open problems for AI-assisted formalization: Part II.

Research note 01, §0 • Kontorovich, via AcadMathSci, Oct 2026 (§2.2) • Bastounis, Circelli & Hansen, arXiv 2610.08144

Part II

# The Lab

# Formalization reaches learning theory

*[ 1.5 min ]  Hook: people in our field are already doing this*

# No verifier, no flywheel

*[ 1 min ]  Why pure math moved first and where our field sits*

# The tools

*[ 1 min ]  One layered map: copilot, project, engine, library*

# A spectrum of human direction

*[ 1 min ]  From fully automatic to human-led; one real case per point*

# Fully automatic

*[ 1 min ]  Case*

# Agents on a human roadmap

*[ 1 min ]  Case*

# AI filling a human blueprint

*[ 1 min ]  Case*

# Human-led, AI-assisted

*[ 1 min ]  Case*

# Running example

*[ 1 min ]  A lemma from the lab's own work, carried through the layers*

# Kernel: reading a Lean file

*[ 2 min ]  What compiles, and what can hide inside it*

# Statement: humans own the statements

*[ 2.5 min ]  Statement review; the part software cannot do*

# Understanding: blueprints and hand-picked lemmas

*[2 min] Human-facing narrative over a machine-built proof*

# What to hand to AI

*[ 1 min ]  The convex hull; what stays human*

# Experiments and applied work

*[ 1 min ]  Code, tests and validation as the certificate*

# Budget

*[ 1 min ]  Compute costs for a student or a lab*

# Checklist

*[ 1 min ]  Each item tied to a layer*

# Closing

# Where this leaves us

*[ 0.5 min ]  Recap: (a) what is coming (Society), (b) how to take part (The Lab)*

# Proof abundance, insight scarcity

*[ 1 min ]  Full circle with the opener and Tao: certificates make proofs abundant; insight is what stays scarce, and it lives in individuals*

# Stay the author

*[ 1 min ]  Individual responsibility: keep control of what is generated, keep your own insights, do not cede your perspective to AI*

# A call to action

*[0.5 min] A few personal commitments for each person in the room*

# Discussion

*[ – min ]  Prompts to open the 15 minutes*

# Corpus and scoring method

*[ – min ]  Details behind the opinion map*

# Backup slides

*[ – min ]  Material for questions*

# References

*[ – min ]  Sources*